Retinoblastoma RNA-seq: Differential Expression and Pathway Analysis

Re-analysis of a DESeq2 results table with pathway enrichment and SQL-based exploration

Summary

Retinoblastoma is a childhood eye cancer driven by loss of the *RB1* tumour suppressor. This report examines a DESeq2 differential expression table (57,773 genes, 19,952 with a valid adjusted p-value) to test whether the data show the biology expected from *RB1* loss. Using thresholds of adjusted p < 0.05 and at least a 2-fold change, **2,168 genes are up and 1,738 are down**. Upregulated genes are enriched for E2F targets, the G2M checkpoint, MYC targets, DNA replication and DNA repair. Downregulated genes are enriched for phototransduction and sensory perception. These results agree with known retinoblastoma biology and with an independent preranked GSEA. The analysis is a re-analysis that recovers known biology, not a new discovery.

1. Background

The RB1 protein normally restrains the E2F family of transcription factors, which turn on genes needed for DNA replication and cell division. When RB1 is lost, E2F activity is released and proliferation genes become highly expressed. MYCN amplification is also reported in aggressive retinoblastoma. Retinoblastoma tumours arise from developing retinal cells, so the normal retinal gene programme, including photoreceptor and phototransduction genes, may be reduced. The question here is whether the expression table shows these two signatures.

2. Data and methods

Step	Detail
Input	DESeq2 results table: baseMean, log2 fold change, lfcSE, Wald statistic, p-value, adjusted p-value (57,773 Ensembl genes)
Filtering	Genes with a valid adjusted p-value (19,952). The rest have zero counts (5,609), were flagged as outliers (about 4,090), or were removed by independent filtering of low-expression genes (about 28,100).
Significance	Adjusted p < 0.05 and absolute log2 fold change at least 1
Annotation	Ensembl ID to HGNC symbol with the MSigDB v7.5.1 mapping (89% of tested genes)
Over-representation	One-sided hypergeometric test, up and down lists separately. Background: tested genes with a symbol (17,744). Gene sets of 10 to 500 genes. Benjamini-Hochberg FDR.
GSEA	Preranked on the Wald statistic, 1,000 permutations, gene sets of 15 to 500 genes (gseapy)
Gene sets	MSigDB Hallmark, KEGG, Reactome, GO Biological Process
Database	Results stored in SQLite and queried with SQL

3. Results

3.1 Differential expression

Of 19,952 tested genes, 3,906 pass both thresholds: 2,168 up and 1,738 down. A further 253 are significant but change by less than 2-fold. The upregulated side is stronger, with more genes and larger fold changes (strongest hits at adjusted p near 10^{-38} and $\log_2$ fold change of 6 to 9). The most significant upregulated genes are mostly mitotic: *DLL3, CKAP2L, NDC80, KIF20A, CENPH, GTSE1, NCAPG*. The most significant downregulated genes include the photoreceptor genes *NRL* and *GRK1*.

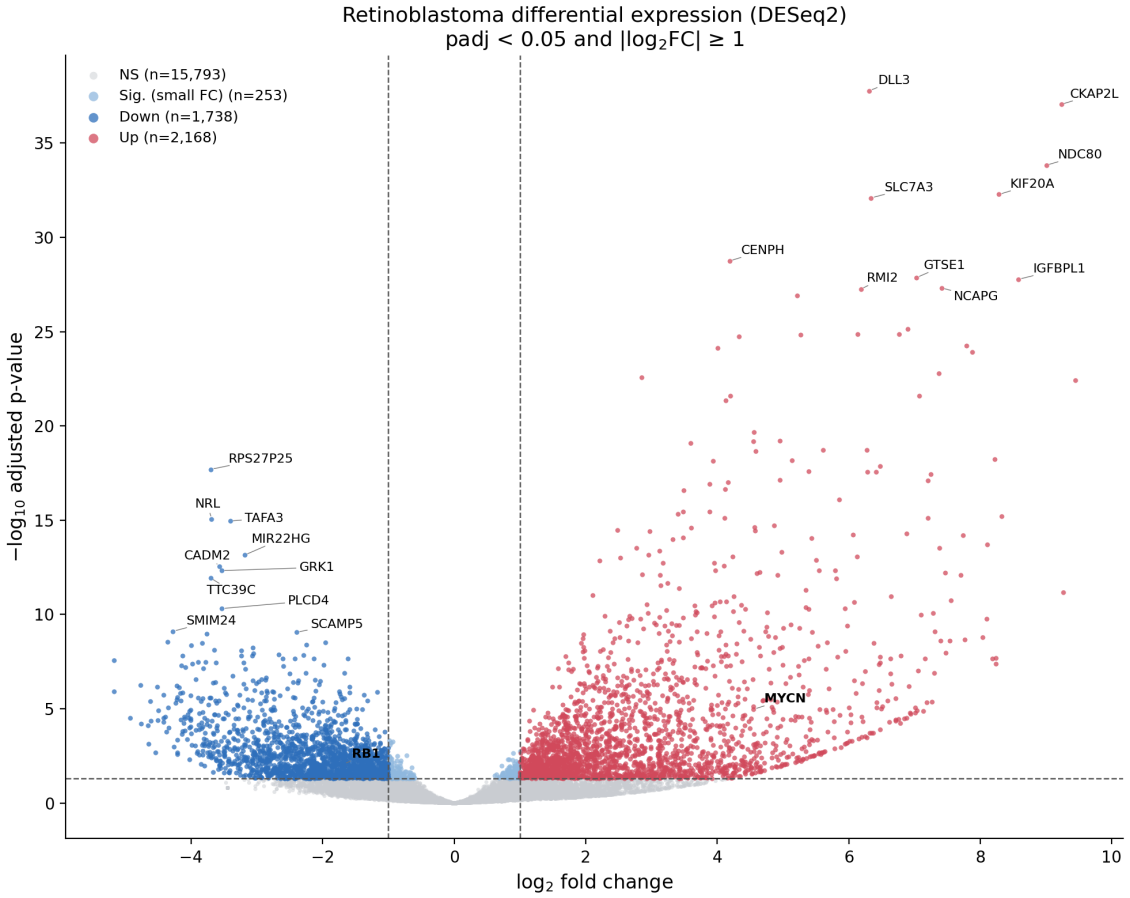


Figure 1. Volcano plot. Red: up (n = 2,168). Blue: down (n = 1,738). Dashed lines: adjusted p = 0.05 and 2-fold change.

3.2 Known retinoblastoma genes

As a sanity check, genes with known roles move in the expected direction.

Gene	Role	log2FC	Adj. p	Direction
RB1	Tumour suppressor	-1.68	8.7e-3	Down
MYCN	Oncogene	+4.55	1.1e-5	Up
MKI67	Proliferation marker	+2.77	3.1e-2	Up
TOP2A	DNA replication	+3.93	8.5e-4	Up
NRL	Rod photoreceptor	-3.69	8.7e-16	Down
GRK1	Phototransduction	-3.54	4.7e-13	Down
RHO	Phototransduction	-4.65	7.4e-4	Down

3.3 Pathway enrichment

Over-representation analysis shows two opposite programmes. Among upregulated genes, **E2F targets** (135 of 198 genes, FDR 3 x 10^{-81}) and the **G2M checkpoint** (116 of 198, FDR 4 x 10^{-59}) dominate, followed by MYC targets, mitotic spindle and DNA repair. KEGG cell cycle (FDR 1.5 x 10^{-15}) and DNA replication (FDR 2.6 x 10^{-15}) are also top terms. Among downregulated genes, Reactome **visual phototransduction** (FDR 7.7 x 10^{-11}) and GO **sensory perception of light stimulus** (FDR 1.6 x 10^{-23}) are strongest.

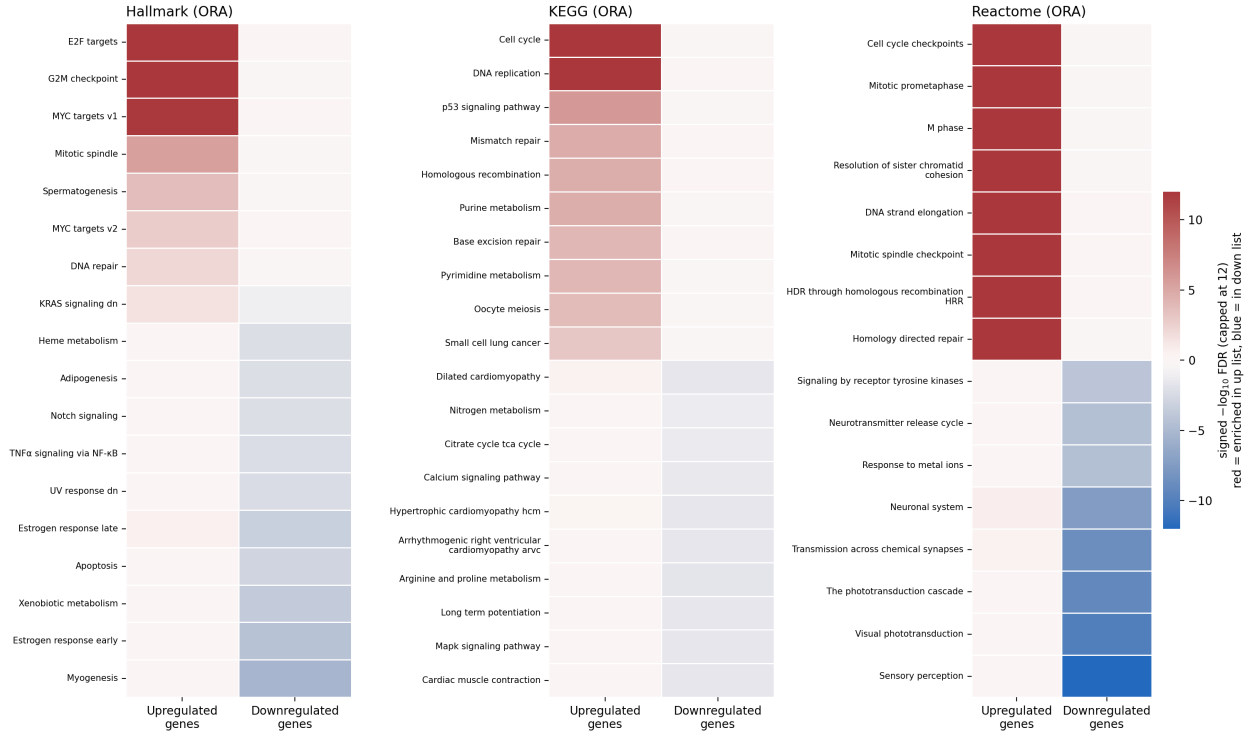


Figure 2. Enrichment of Hallmark, KEGG and Reactome terms for the up and down gene lists (signed -log10 FDR, capped at 12).

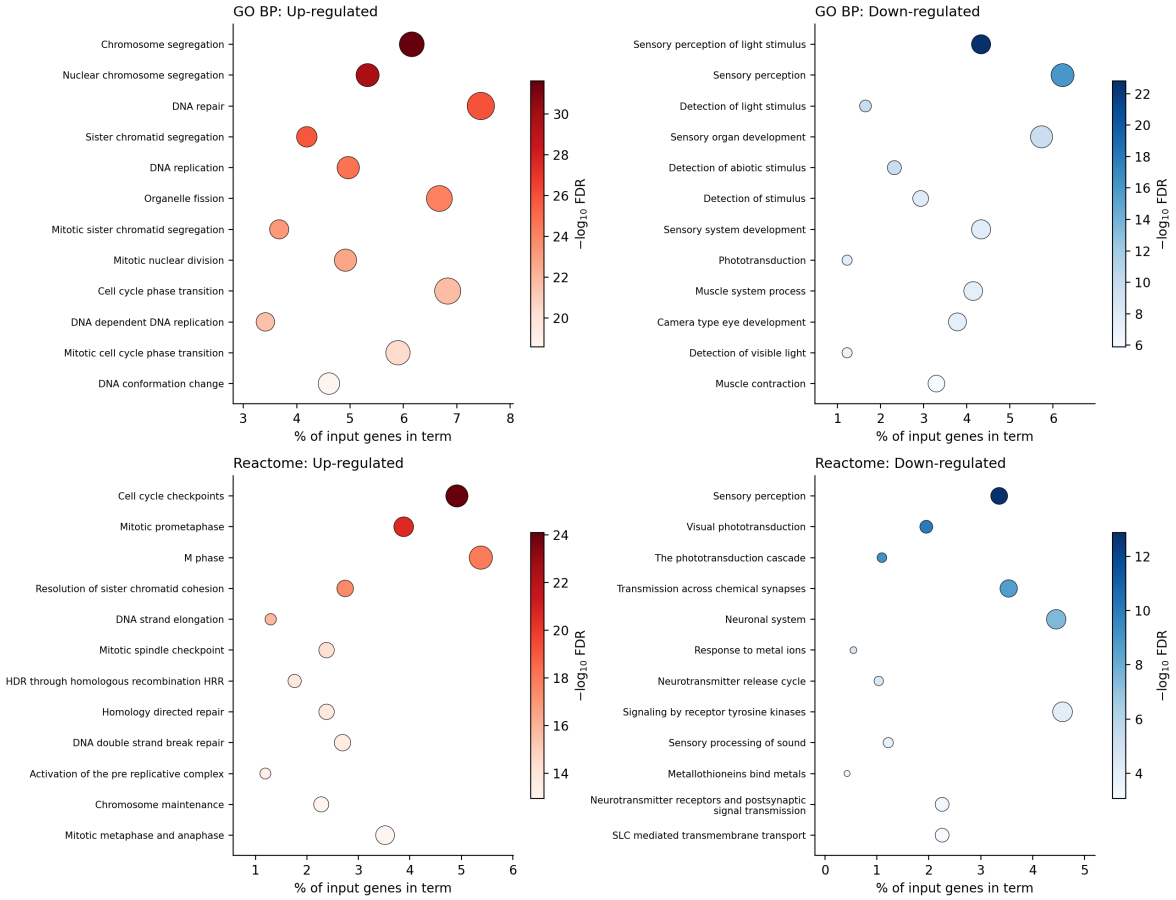


Figure 3. Top GO Biological Process and Reactome terms. Dot size: number of genes in the term. Colour: -log10 FDR.

3.4 Preranked GSEA

GSEA uses all genes ranked by the Wald statistic and does not depend on a threshold. It agrees with the over-representation result: E2F targets (NES 3.54), G2M checkpoint (3.35) and MYC targets V1 (2.75) are positively enriched, whereas the phototransduction cascade (NES -2.55) and oxidative phosphorylation (Hallmark NES -2.16) are negatively enriched. For Hallmark, 26 sets are significant at FDR below 0.05 (8 positive, 18 negative).

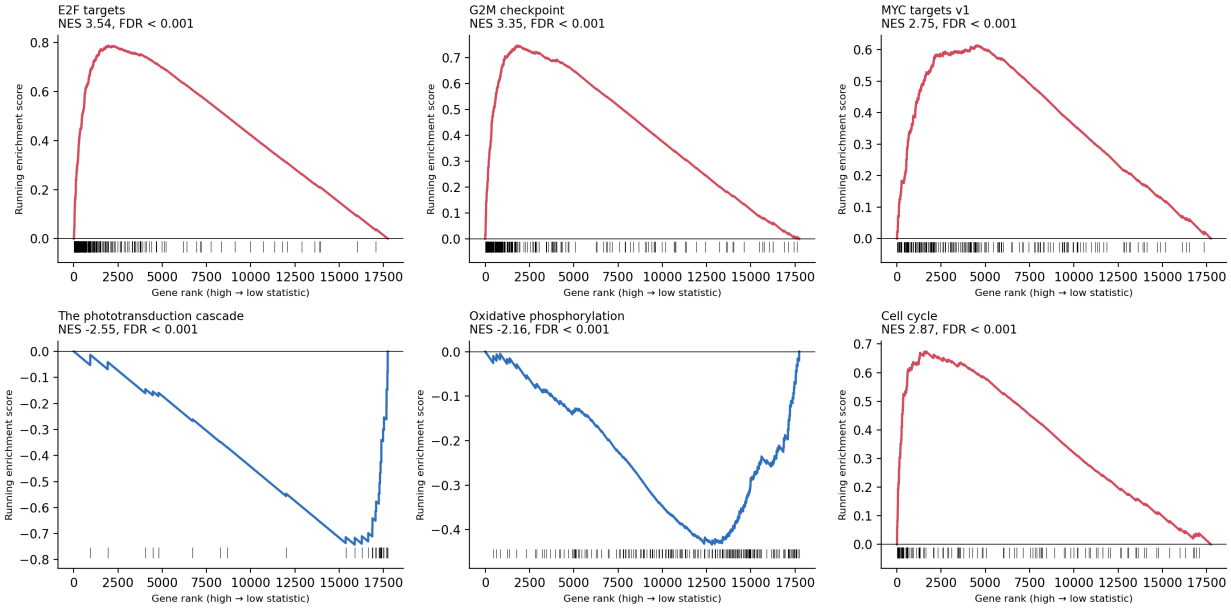


Figure 4. GSEA running enrichment scores. Red curves: positively enriched. Blue curves: negatively enriched.

3.5 Genes behind the pathways

The heatmap lists the most strongly changed genes in each key pathway. Cell-cycle genes such as *NDC80, BUB1B, E2F1, MCM2* and *MCM6* are strongly up, while the visual-cycle genes *RHO, GNAT1, PDE6B* and *GRK1* are down.

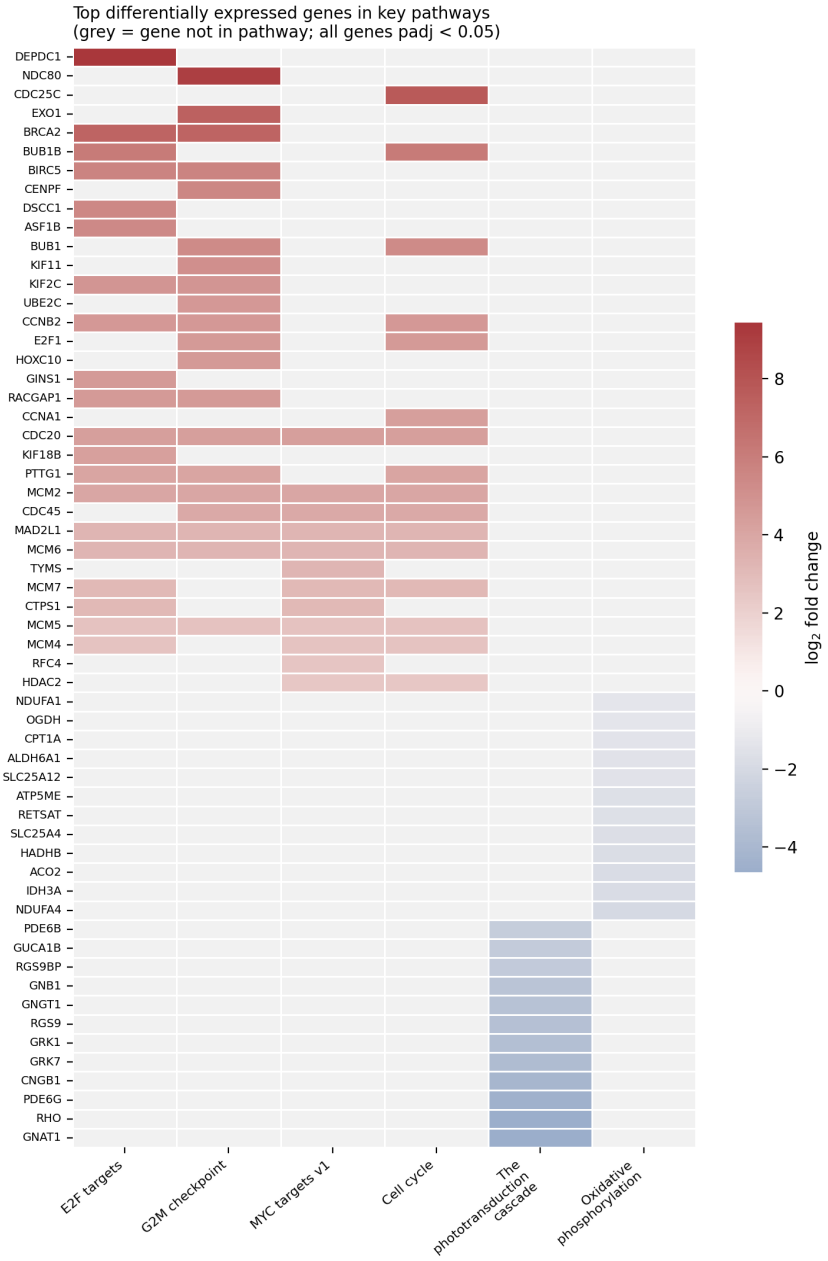


Figure 5. log2 fold change of the strongest genes in selected pathways (grey: gene not in the pathway; all genes adjusted p < 0.05).

3.6 Quality control

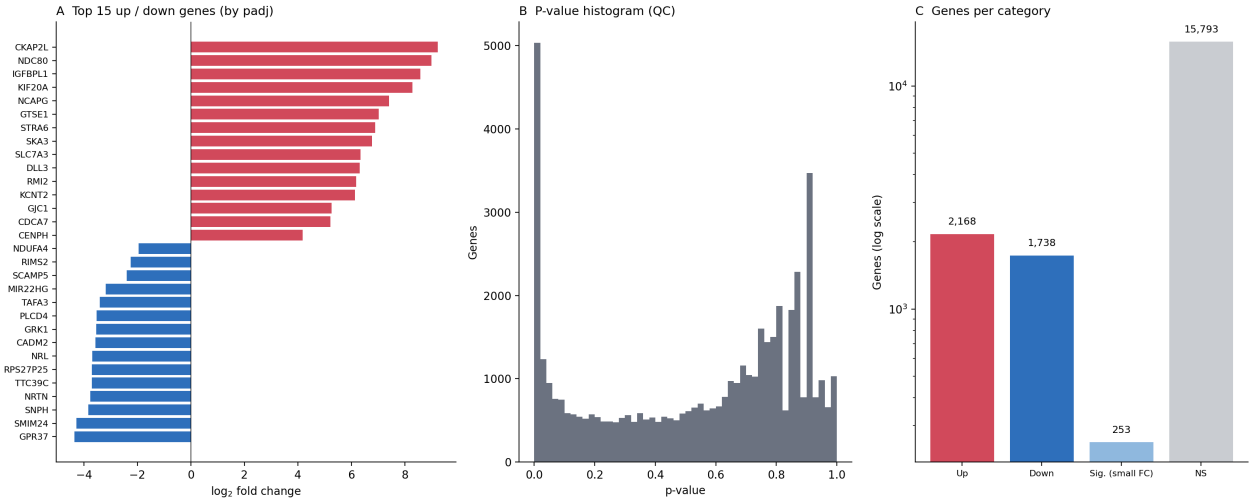


Figure 6. (A) Top 15 up and down genes by adjusted p-value. (B) P-value histogram. (C) Genes per category (log scale).

The p-value histogram shows a strong spike near zero, as expected when many genes are truly differentially expressed. The pile-up towards p = 1 comes from low-count genes with discrete counts; DESeq2 independent filtering removes most of these before adjustment.

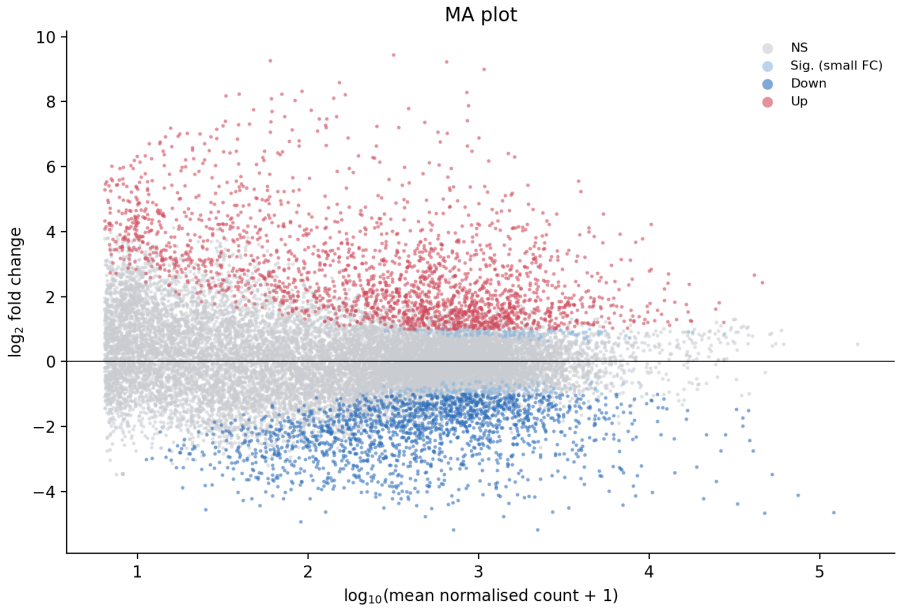


Figure 7. MA plot: mean expression against log2 fold change.

4. SQL analysis

The results were loaded into a SQLite database with four tables (`genes`, `enrichment_ora`, `enrichment_gsea`, `hallmark_membership`). Seven queries explore them with joins, CTEs, window functions, aggregation and binning. Full outputs are in `sql/query_results.md`. One example finds the Hallmark pathways with the highest mean fold change:

```
SELECT m.pathway, COUNT(*) AS genes_tested,
       SUM(g.status = 'Up') AS n_up,
       ROUND(AVG(g.log2fc), 2) AS mean_log2fc
FROM hallmark_membership m
JOIN genes g ON g.symbol = m.symbol
GROUP BY m.pathway HAVING COUNT(*) >= 30
ORDER BY mean_log2fc DESC LIMIT 5;
```

Pathway	Genes tested	Up	Mean log2FC
E2F targets	198	135	+1.99
G2M checkpoint	198	116	+1.77
MYC targets V2	56	16	+0.94
MYC targets V1	199	60	+0.80
Spermatogenesis	101	26	+0.73

5. Discussion

The data show a coherent picture. Loss of RB1 releases E2F, and E2F target genes are the single most enriched set in both methods. MYCN is strongly up and MYC target sets are enriched, which fits the role of MYC family signalling in aggressive retinoblastoma. In parallel, genes that define specialised retinal cells, especially photoreceptor phototransduction genes, are lower in the tumour. The oxidative phosphorylation signal in GSEA is also lower, which may reflect the loss of metabolically active photoreceptors. The result is a proliferative, DNA-replication-heavy profile with reduced retinal identity.

6. Limitations

- **Cell composition:** the photoreceptor signal may partly reflect that normal retina is rich in photoreceptors, not only tumour-intrinsic repression. Cell-type deconvolution or single-cell data would be needed to separate these.
- **Contrast direction** is not stored in the input file. Positive log2 fold change is assumed to mean higher in retinoblastoma.
- **Annotation:** 11% of tested genes had no gene symbol and were excluded from enrichment.
- **Noisy fold changes:** median lfcSE is about 0.6. Applying lfcShrink (apeglm) is advisable before ranking genes by effect size.
- **No sample-level analysis:** PCA and sample heatmaps need the raw count matrix, which was not available.
- **Non-specific hits:** some downregulated Hallmark sets (for example estrogen response, myogenesis) are broad signatures and should be interpreted cautiously.
- **Gene-set version:** MSigDB v7.5.1; newer releases differ slightly.

7. How this analysis was made

The Python analysis scripts and SQL queries were generated with AI assistance (Claude, Anthropic). The project owner framed the question, supplied the DESeq2 table, reviewed the outputs, checked them against known retinoblastoma biology and wrote the interpretation. The DESeq2 statistics were produced before this project. Code should be re-run and verified before reuse.

References

- Love MI, Huber W, Anders S (2014). Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology* 15:550.
- Subramanian A et al. (2005). Gene set enrichment analysis. *PNAS* 102:15545-15550.
- Liberzon A et al. (2015). The Molecular Signatures Database Hallmark gene set collection. *Cell Systems* 1:417-425.
- Fang Z, Liu X, Peltz G (2023). GSEApy: a comprehensive package for performing gene set enrichment analysis in Python. *Bioinformatics* 39:btac757.
- Benjamini Y, Hochberg Y (1995). Controlling the false discovery rate. *J R Stat Soc B* 57:289-300.